THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

OBJECT ORIENTED PROGRAMMING

ASSIGNMENT NO.6

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Q1. Implement a class Library that contains an array of Book objects. The Book class should have attributes like title, author, and ISBN.

CODE:

```
#include <iostream>
using namespace std;

class Book {
public:
  string title;
  string
  author;
  string
  ISBN;
};

class Library {
  Book
  books[100];
  int count;

public:
  Library() {
    count = 0;
  }
  bool addNewBook(string &title, string &author, string
    &ISBN) { if (count >= 100) return false;
    books[count].title = title; books[count].author =
    author; books[count].ISBN = ISBN; count++;
```

```
    return true;
  }

  bool removeBooks(string &ISBN);

  void displayDetails() {
    if (count == 0) {
      cout << "No books in
    library\n"; return; }
    for (int i = 0; i < count; i++) {
      cout << "\nBook " << i + 1 << ":\n";
      cout << "Title: " << books[i].title <<
      endl; cout << "Author: " <<
      books[i].author << endl; cout <<
      "ISBN: " << books[i].ISBN << endl;
    }
  }
};

bool        Library::removeBooks(string
&ISBN) {
  for (int i = 0; i < count; i++) {
    if (books[i].ISBN == ISBN)
      { for (int j = i; j < count - 1;
      j++) {
        books[j] = books[j + 1];
      } count--
      ; return
      true;
```

```
    } }
  return
  false;
}

int    main()    {
  Library lib;
  string t1 = "Maths", a1 = "Rachit", i1
  = "101";
  string t2 = "Data Structures", a2 = "Siya", i2
  = "102"; string t3 = "Analog", a3 = "Arun",
  i3 = "103"; string t4 = "DBMS", a4 =
  "Neha", i4 = "104"; string t5 = "OS", a5 =
  "Kirti", i5 = "105";

  lib.addNewBook(t1,
  a1,                    i1);
  lib.addNewBook(t2,
  a2,                    i2);
  lib.addNewBook(t3,
  a3,                    i3);
  lib.addNewBook(t4,
  a4,                    i4);
  lib.addNewBook(t5,
  a5, i5);

cout << "\n Books after adding \n";
lib.displayDetails();          string
removeISBN     =     "103";     if
(lib.removeBooks(removeISBN))
```

```
    cout << "\nBook removed
successfully\n"; else
    cout << "\nBook not found\n";
cout << "\n Books after removal
\n"; lib.displayDetails(); return 0;
}
```

OUTPUT:

```
PS D:\downlod\OOPS\Assignment6> cd "d:\downlod\OOPS\Assignment6\" ; if (

     Books after adding

Book 1:
Title: Maths
Author: Rachit
ISBN: 101

Book 2:
Title: Data Structures
Author: Siya
ISBN: 102

Book 3:
Title: Analog
Author: Arun
ISBN: 103

Book 4:
Title: DBMS
Author: Neha
ISBN: 104

Book 5:
Title: OS
Author: Kirti
ISBN: 105

Book removed successfully

     Books after removal

Book 1:
Title: Maths
Author: Rachit
ISBN: 101

Book 2:
Title: Data Structures
Author: Siya
ISBN: 102

Book 3:
Title: DBMS
Author: Neha
ISBN: 104

Book 4:
Title: OS
Author: Kirti
ISBN: 105
PS D:\downlod\OOPS\Assignment6>
```

Q2. Implement the question 1 using constructor. Do the following modification:

a. **Add default constructor, parameterized constructor, and copy constructor to add the details of the book using the same parameters, i.e., title, author, and ISBN.**

b. **Use the "this" pointer instead of scope resolution operator to access the data members of the class inside the constructor and function definition.**

c. **Create function bool removeBooks(ISBN) same as the question 1.**

d. **displayDetails() same as question 1.**

e. **Create array of objects using 1. Initializer list, 2. Dynamic initialization**

CODE:

```
#include <iostream>
using namespace std;


class Book {
public:
  string title;
  string
  author;
  string
  ISBN;

  Book() {
    title = "";
    author = "";
    ISBN = "";
  }
  Book(string title, string author, string
    ISBN) { this->title = title; this-
    >author = author; this->ISBN =
    ISBN;
  }
```

```
  Book(const Book
    &b) { this->title =
    b.title; this->author
    = b.author; this-
    >ISBN = b.ISBN;
  }
};

class Library {
  Book
  books[100];
  int count;

public:
  Library() {
    count = 0;
  }
  bool addNewBook(string &title, string &author, string &ISBN) {
    if (count >= 100) return false;
    books[count] = Book(title, author,
    ISBN); count++; return true;
  }

  bool removeBooks(string
    &ISBN) { for (int i = 0; i <
    count; i++) {
      if (books[i].ISBN == ISBN)
        { for (int j = i; j < count - 1;
        j++) {
          books[j] = books[j + 1];
```

```
            } count--
            ; return
            true;
        }
    }    return
    false;
  }

  void displayDetails() {
    if (count == 0) {
      cout << "No
    books\n"; return; }
    for (int i = 0; i < count; i++) {
      cout << "\nBook " << i + 1 << endl;
      cout << "Title: " << books[i].title <<
      endl;  cout  <<  "Author:  "  <<
      books[i].author  <<  endl;  cout  <<
      "ISBN: " << books[i].ISBN << endl;
    }
  }
};

int main() {
  Book arr1[3] = {
    Book("C++", "Ram", "101"),
    Book("DSA", "Shyam", "102"),
    Book("OS", "Amit", "103")
  };
```

```
cout << "Initializer list array:\n"; for (int i = 0; i < 3; i++) {
cout << arr1[i].title << " " << arr1[i].author << " " << arr1[i].ISBN << endl;
}

int n = 2;
Book *arr2 = new Book[n]; arr2[0] = Book("DBMS", "Neha", "104");
arr2[1] = Book("Physics", "Kiran", "105");

cout << "\nDynamic array:\n"; for (int i = 0; i < n; i++) {
cout << arr2[i].title << " " << arr2[i].author << " " << arr2[i].ISBN << endl;
}

delete[] arr2;
Library lib;
string t1="Math", a1="A", i1="201"; string t2="Physics", a2="B", i2="202"; string t3="Chemistry", a3="C", i3="203"; string t4="Biology", a4="D", i4="204"; string t5="English", a5="E", i5="205";
```

```
PS D:\download\OOPS\Assignment6> cd "d:\download\OOPS\Assignment6\"
Initializer list array:
C++ Ram 101
DSA Shyam 102
OS Amit 103

Dynamic array:
DBMS Neha 104
Physics Kiran 105

Library books:

Book 1
Title: Math
Author: A
ISBN: 201

Book 2
Title: Physics
Author: B
ISBN: 202

Book 3
Title: Chemistry
Author: C
ISBN: 203

Book 4
Title: Biology
Author: D
ISBN: 204

Book 5
Title: English
Author: E
ISBN: 205
```

```
  lib.addNewBook(t1,a1,i1);
  lib.addNewBook(t2,a2,i2);
  lib.addNewBook(t3,a3,i3);
  lib.addNewBook(t4,a4,i4);
  lib.addNewBook(t5,a5,i5);

  cout << "\nLibrary
  books:\n";
  lib.displayDetails();
  string rem = "203";
  lib.removeBooks(rem);
 cout << "\n After removing one book: \n";
  lib.displayDetails();
 return 0;
}
```

```
     After removing one book:

Book 1
Title: Math
Author: A
ISBN: 201

Book 2
Title: Physics
Author: B
ISBN: 202

Book 3
Title: Biology
Author: D
ISBN: 204

Book 4
Title: English
Author: E
ISBN: 205
PS D:\downlod\OOPS\Assignment6>
```

Q3. Create a class Account with attributes account number (of const long type), transaction ID (of long type), transaction type (credit/debit) (of string type), and balance (of double type).

CODE:

```
#include <iostream>
using namespace std;

class Account {
private:
  const long
  accNo; long
  transID; string
  transType;
```

```
  double
  balance;

public:
  Account(long accNo, long transID, string transType, double balance)
    :
    accNo(accN
    o)
  {
    this->transID = transID;
    this->transType =
    transType; this-
    >balance = balance;
  }
  void display() {
    cout << "Account No: " << accNo << endl;
    cout << "Transaction ID: " << transID <<
    endl; cout << "Transaction Type: " <<
    transType << endl; cout << "Balance: " <<
    balance << endl;
  }
};

int main() {
  Account a1(1001, 5001, "credit", 15000.5);
  a1.display();
  return 0;
}
```

```
PS D:\downlod\OOPS\Assignment6> cd "d:\downlod\OOPS\Assignment6\"
Account No: 1001
Transaction ID: 5001
Transaction Type: credit
Balance: 15000.5
PS D:\downlod\OOPS\Assignment6>
```

Q4. Write a program to add data objects of two different classes using friend functions.

CODE:

```
#include   <iostream>
using namespace std;

class B;
class  A  {
   int x;
public:
   A(int x) { this-
      >x = x;
   }
   friend int add(A, B);
};


class  B  {
   int y;
public:
   B(int y) { this-
      >y = y;
   }
   friend int add(A, B);
};


int add(A obj1, B obj2) {
   return obj1.x + obj2.y;
}
int main() {
   A
   obj1(10);
```

```
    B
    obj2(20);
    int result = add(obj1, obj2);
    cout << "Sum = " << result << endl;


    return 0;
}
```

OUTPUT:

```
Balance: 15000.5
PS D:\downlod\OOPS\Assignment6> cd "d:\downlod\OOPS\Assignment6\" ; if
 .\tempCodeRunnerFile }
Sum = 30
PS D:\downlod\OOPS\Assignment6>
```

Q5. Define a class named Complex with properties (real and imaginary) and methods as per following details.

a. Parameterized constructor and copy constructor to initialize object values.

b. void display () to display complex number.

c. void sum (Complex, Complex) to add two complex numbers (objects of Complex class). Implement this function as the friend function.

CODE:

```
#include   <iostream>
using namespace std;


class   Complex
   {   int    real,
   imag;
public:
   Complex(int r, int
      i)  {  real  =  r;
      imag = i;
   }
```

```
    Complex(const  Complex
      &c) { real = c.real; imag
      = c.imag;
    }
    void display() {
cout << real << " + " << imag << "i" << endl;
    }
    friend void sum(Complex, Complex);
};

void sum(Complex c1, Complex c2) { int r =
    c1.real + c2.real; int i = c1.imag + c2.imag;
    cout << "Sum = " << r << " + " << i << "i"
    << endl;
}

int main() {
    Complex c1(2, 3);
    Complex     c2(4,     5);
    Complex  c3(c1);  cout
    <<  "First  complex:  ";
    c1.display();   cout   <<
    "Second   complex:   ";
    c2.display();   cout   <<
    "Copied   complex:   ";
    c3.display();
    sum(c1, c2);

    return 0;
}
```

```
PS D:\downlod\OOPS\Assignment6> cd "d:\downlod\OOPS\Assignment6\" ;
 .\tempCodeRunnerFile }
First complex: 2 + 3i
Second complex: 4 + 5i
Copied complex: 2 + 3i
Sum = 6 + 8i
PS D:\downlod\OOPS\Assignment6>
```